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10. Thiosulfate ions react with acid according to the equation shown.

$$S_2O_3^{2-}(aq) + 2H^+(aq) \longrightarrow S(s) + SO_2(g) + H_2O(l)$$

A student investigates the effect of changing the concentration of thiosulfate ions on the rate of the reaction.

(a) Describe how the student could prepare 250 cm^3 of 0.10 $mol\,dm^{-3}$ sodium thiosulfate solution to use in this experiment. The relative formula mass (M_r) of sodium thiosulfate, $Na_2S_2O_3.5H_2O$, is 248.3.

You should include the apparatus needed and any relevant masses and volumes. [4]

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(b) The student mixes known volumes of aqueous 0.10 mol dm^{-3} sodium thiosulfate and hydrochloric acid in a beaker. He places the beaker on a cross drawn on a piece of paper and measures the time taken for the cross to be obscured.

(i) State what causes the cross to become obscured. [1]

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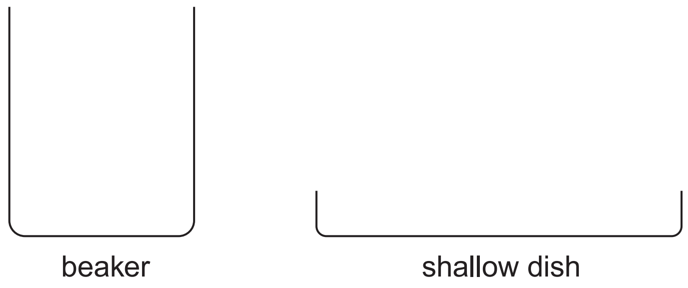
(ii) The diagram shows a beaker and a shallow dish.

If the student replaced the beaker with a shallow dish, state the effect this would have on the time taken to obscure the cross. Explain your answer.

Assume that the student used the same volumes and concentrations in each experiment. [1]



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(c) The student plans to change the concentration of $S_2O_3^{2-}$(aq) by reducing the volume of aqueous sodium thiosulfate and adding water to keep the total volume constant.

The data obtained by the student for one experiment is shown in the table.

Volume $S_2O_3^{2-}$(aq) / cm^3	Volume H_2O / cm^3	Volume HCl(aq) / cm^3	Time / s
40	0	10	50

(i) **Complete the table** to show suitable volumes of thiosulfate, water and acid that the student could use to investigate the effect of changing concentration of $S_2O_3^{2-}$(aq) on the rate of the reaction. [2]

(ii) The rate of reaction can be calculated using rate = $\frac{1000}{\text{time}}$

State the unit of rate in this reaction. [1]

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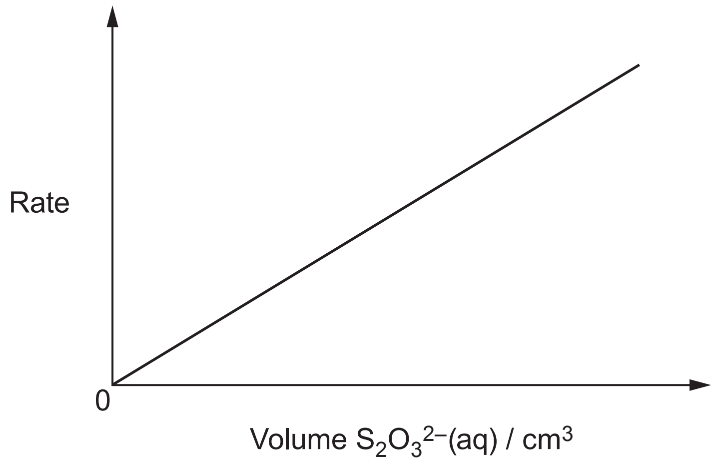
(iii) The student calculates the rate of reaction using different volumes of $S_2O_3^{2-}(aq)$ and plots a graph. The graph is shown.



State what can be deduced from the shape of this graph. [1]

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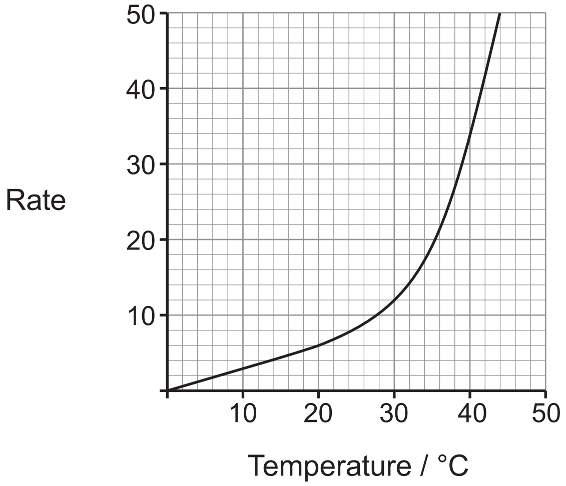
(d) Another student uses a similar experiment to investigate the effect of changing the temperature on the rate of this reaction. He draws a graph of his data.



(i) Use the graph to determine the time taken for the cross to be obscured at a temperature of 30 °C. Show clearly how you obtained your answer. [2]

$$\text{rate} = \frac{1000}{\text{time}}$$

Time = ... s

(ii) Some textbooks state that a 10 °C temperature rise approximately doubles the rate of many reactions. Use the graph to find to what extent this is true for this reaction. [2]

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(iii) Explain why the rate of the reaction is affected by an increase in temperature. [2]

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